

- Q.26 Explain slug and strip lay out.
- Q.27 What is shut height & die cushion.
- Q.28 Explain the principle of maximum and minimum material conditions in limit gauge design.
- Q.29 What are the main functions are features of jig and fixtures?
- Q.30 What are the two main classifications of integrated dies?
- Q.31 What factors are considered when selecting a press for a specific application?
- Q.32 Explain the significance of personal safety in die maintenance activities.
- Q.33 What are the various steps performed in press working operations?
- Q.34 What are design parameters for strip layout?
- Q.35 Explain the principle of maximum and minimum material condition in limit gauge design.

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 What is stripping force? Explain important factors which affect stripping force.
- Q.37 Explain the role of jigs and fixtures in the manufacturing process and discuss their applications in improving productivity and quality. Also Explain the process of analysing steps in designing jigs and fixture.
- Q.38 Explain the various elements and parts in a typical press tool with neat and clear diagram.

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Roll No.

2nd Year /Advance Diploma in Tool and Die Making Subject:- Tool Design Theory-I (Press Tool, Jigs & Fixtures)

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 _____ is a metal-cutting process used to cut vertically down and perpendicular to the surface on sheet-metal from the edge of a work-piece
- a) Perforating b) Parting
c) Blanking d) Notching
- Q.2 Shearing the sheet into two or more piece is known as?
- a) Parting b) Punching
c) Blanking d) Notching
- Q.3 In blanking operation, the clearance is provided on the _____.
- a) Die b) Punch
c) In both d) None of the above
- Q.4 It is a cutting operation for producing desired shape and size hole in the product is known as a punching?
- a) True b) False
- Q.5 As the thickness of the sheet is increased the clearance needed will also?
- a) Increase b) Decrease
c) No effect
d) First decreases and then increase

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- Q.6 With the use of jigs and fixture total cost of production.
- Increases
 - Decreases
 - Remains same
 - Jigs are not used in any production process
- Q.7 Fixture is a device of which _____?
- holds and lockets workpiece and guide the cutting tool.
 - Hold and position the workpiece but does not guide the tool.
 - is made lighter for quicker handling
 - is used for holding workpiece in drilling, reaming and tapping operation
- Q.8 The purpose of jig and fixture is to?
- increase production rate
 - increase machining accuracy
 - facilitate interchangeable manufacturer
 - all of these
- Q.9 Which among the following statements is not correct?
- jig is used for planning operation
 - jig is used for drilling operation
 - jig is used for indexing operation
 - jig is used for boring operation
- Q.10 Drawing operations are essential for forming:
- Cups
 - Bolts
 - Spring
 - None of the above

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Drawing operations are essential for forming _____
- Q.12 The 3-2-1 pin concept of location in jigs and fixtures refers to a specific method of locating work _____.
- Q.13 What is spring back?
- Q.14 The concept of _____ involves the tendency of metal to return to its original shape after bending or forming.
- Q.15 Define integrated operation.
- Q.16 Write function of press tool.
- Q.17 Press tools are generally _____ working process.
- Q.18 Define tolerance.
- Q.19 The space per side between the punch and die is called _____
- Q.20 What is plain limit gauge.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Explain the concept of press tools in the context of mass production.
- Q.22 Explain the main components of a press tool and give their functions.
- Q.23 Enlist the factor to be considered for the selection of a die.
- Q.24 Explain the concept of the 3-2-1 pin principle of location in jigs and fixtures.
- Q.25 What are the key design parameters considered in the development of limit gauges?